

## Assignment 9

### Incident report

In late 2025 and early 2026, users of the social media platform **X** began using the AI assistant **Grok** to edit photographs of real people, particularly women, by making them appear partially undressed or in revealing clothing without their consent. The manipulated images were generated directly through Grok's image-editing feature and were often shared publicly on X. Reports also documented cases involving celebrities, journalists, and, in some instances, children.

The altered images were created by X users through prompts submitted to Grok, which generated edited versions of existing photographs. Following public criticism and regulatory scrutiny, xAI and X introduced restrictions on the image-editing feature, including limiting access and adding safeguards in some jurisdictions. Investigations by regulators in several countries followed, while reports indicated that some image-editing capabilities remained available through other Grok interfaces.

### 2 Agents

| Agent             | Role                                                           |
|-------------------|----------------------------------------------------------------|
| End User          | Uploads, prompts Grok and selects images                       |
| AI system (Grok)  | Interprets prompts and generates new versions of images        |
| AI developer X-AI | Designs, trains and deploys Grok's image generation capability |
| Platform (X)      | Hosts Grok, distributes images, manages sharing and moderates  |

### 3 Mediation Pathways

Agent 1: End User (Local/Remote when the image is already on the platform)

End User → Uploads another person's photo → Prompts Grok to "undress" or sexualise the image → Shares generated image on X → Non-consensual image spreads

Outcome: The manipulated image becomes publicly visible, potentially reaching large audiences.

Agent 2: AI System (Grok) (Remote)

Grok → Receives image and prompt → Generates edited image → Returns output to user → Image is redistributed

Agent 3: X (Remote)

X → Hosts generated image → Displays it to other users → Enables reposting and engagement → Wider public exposure

### 3.1 Harm Caused

Victims: Loss of privacy, reputational damage, humiliation, emotional distress, and loss of control over their own image. Journalists and public figures reported feeling dehumanized after manipulated images of them circulated online.

Platform and society: Increased circulation of non-consensual sexual deepfakes, erosion of trust in digital images, and regulatory investigations into X and xAI.

## Part 4 – EDOCA Assessment of Critical Transitions

### 4.1 Transition 1: User Prompt → Grok Generates a Manipulated Image

#### 1. Transition ( $S \rightarrow V \rightarrow D$ )

Source (S): A user uploads a real person's photo and submits a prompt requesting an "undressed" version.

Vector (V): Grok interprets the prompt and generates an edited image using its image-generation model.

Destination (D): A manipulated image is returned to the user.

Space transition: Personal → Remote; Physical → Virtual; Symbolic → Computational.

#### 2. EDOCA

Effort/Energy: Low effort is required from the user (uploading a photo and writing a prompt). Computational resources are provided by xAI's cloud infrastructure.

Observability: The user sees only the final image. The model's internal reasoning and safety filtering are largely opaque to users and victims, while xAI has greater visibility into system operations.

Control: The user controls the prompt, while xAI controls the model, training data, and safety guardrails. The victim has no control over the process.

Authority: xAI has the authority to enable, restrict, or disable image-generation features. Users can only use the system within those limits.

Distortion/Fidelity: The AI transforms an authentic photograph into a fabricated image, significantly reducing fidelity and creating misleading content.

#### 3. Synthesis

This transition creates the primary harm because it converts a real person's identity into a false sexualised image. Although the AI acts according to its design, insufficient guardrails allowed harmful outputs. Responsibility therefore remains with both the user who initiated the request and xAI, which designed and deployed the system.

#### 4. Signal and State integrity

Signal degradation / distortion: The original photograph is transformed into a fabricated sexualized image, reducing the fidelity of the person's real appearance.

State divergence: Grok treats the prompt as an acceptable image-editing request, while the real-world state is that the person did not consent to the manipulation. The system's internal representation does not reflect this reality.

Symbolic failure: The model interprets text prompts and image content without fully understanding social context, consent, or potential harm, leading to an inappropriate representation.

State failure: The system generates and returns an image as if the request were valid, despite violating the subject's rights and the intended purpose of safety guardrails.

Bias: Women, public figures, and other frequently photographed individuals are disproportionately targeted, resulting in systematic harm to these groups.

## Transition 2: Generated Image $\rightarrow$ Distribution on X

### 1. Transition ( $S \rightarrow V \rightarrow D$ )

Source (S): AI-generated manipulated image.

Vector (V): X hosts, recommends, and enables sharing of the image.

Destination (D): Other users view, repost, and further distribute the content.

Space transition: Remote  $\rightarrow$  Remote; Computational  $\rightarrow$  Social.

### 2. EDOCA

Effort/Energy: Minimal effort is needed to post or repost content. X provides the platform infrastructure and recommendation systems.

Observability: The image is publicly visible, but recommendation algorithms and moderation decisions are only partially visible to users.

Control: Users decide to post and share content, while X controls content moderation, recommendation algorithms, and removal mechanisms. Victims have limited control once content spreads.

Authority: X has the authority to remove content, suspend accounts, or change moderation policies.

Distortion/Fidelity: The manipulated image is amplified through sharing, increasing its reach and making it more likely to be perceived as authentic.

### 3. Synthesis

This transition amplifies harm by rapidly distributing manipulated images beyond the original creator. Responsibility is shared between users who share the content and the platform that determines how widely it is promoted and how quickly it is removed.

#### 4. Signal and State integrity

Signal degradation / distortion: As the manipulated image is shared, viewers may mistake it for a genuine photograph, further degrading the accuracy of the original information.

State divergence: The platform treats the image as ordinary user content until moderation occurs, while the actual state is that it is non-consensual manipulated media causing harm.

Symbolic failure: The platform may fail to correctly identify or label the image as AI-generated or manipulated, allowing misleading representations to spread.

State failure: Moderation systems may not remove harmful images quickly enough, creating a mismatch between platform policies and what users actually experience.

Bias: Harm is concentrated on individuals who are more likely to become targets of image manipulation, particularly women, journalists, celebrities, and other public figures.

#### Overall Synthesis

The two transitions show how responsibility becomes distributed across multiple actors. The user initiates the harmful request, Grok transforms the image, and X amplifies its reach. The AI does not intentionally "decide" to cause harm, but inadequate safeguards allowed harmful outputs to be generated. While the vendor may describe these outcomes as unintended, responsibility does not disappear—it remains shared between the user, xAI as the developer, and X as the platform responsible for moderation and oversight.

#### Part 6: Capability vs Intent and consequences

##### Step 1: Stated Intent

Grok was designed by xAI as a generative AI assistant capable of answering questions and creating or editing images based on user prompts. Its image-generation features were intended to support creative expression and make AI image editing widely accessible. Publicly, the system was presented as a helpful and innovative AI tool, with safety measures intended to prevent harmful or abusive outputs.

##### Step 2: Actual Capability

In practice, Grok could generate manipulated images of real people, including sexualised or "undressed" versions, when prompted by users. Reports showed that the system's safeguards were insufficient to prevent non-consensual image manipulation. The risks of AI-generated deepfakes and non-consensual intimate imagery were already well documented in academic research and public discussions before this incident, making these failure modes known rather than unexpected.

##### Step 3: Analysis of the Consequence

The harm is best described as foreseeable but not prevented, rather than genuinely unintended. The capability to generate realistic edited images was intentional, and the risks of misuse had already been widely recognised. As shown in the EDOCA analysis, xAI had control over model design and safety guardrails, while X had authority over content moderation and distribution. Although neither actor intended users to create abusive images, both had opportunities to reduce or prevent the harm through stronger safeguards and moderation. The case also reflects plausible deniability to some extent: users initiated the harmful prompts, the AI generated the images, and the platform distributed them, spreading responsibility across multiple actors even though the overall risk was predictable.

## Part 7: Responsibility

Ultimately responsibility is spread across the two actors, namely the X-User and the xAI. Although the prompts were sent by users, xAI could have used its authority over moderation to deny requests of this nature.

## Part 8: Reflection

Through reconstructing the mediation pathways I had an easier time understanding what actions are harmful and who they are a responsibility of. Through that I was able to determine responsibility of the agents involved.